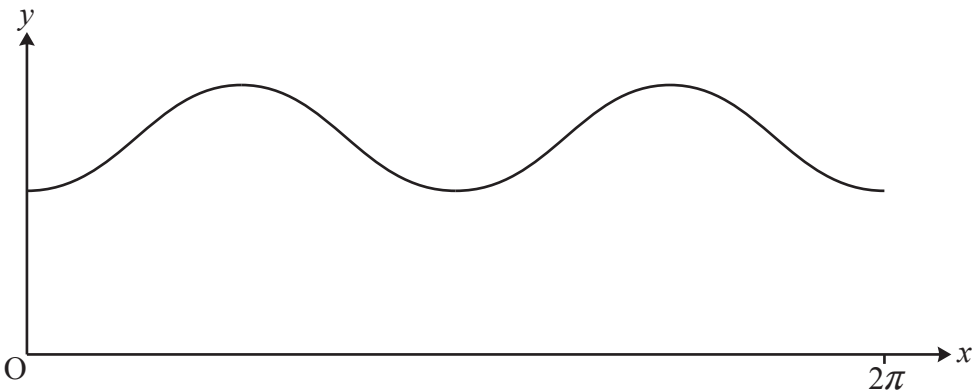
10 The diagram shows the graph of $y = 1.5 + \sin^2 x$ for $0 \leqslant x \leqslant 2\pi$.



(a) Show that the equation of the graph can be written in the form $y = a - b\cos 2x$ where a and b are constants to be determined. **[2]**

(b) Write down the period of the function $1.5 + \sin^2 x$. **[1]**

(c) Determine the x-coordinates of the points of intersection of the graph of $y = 1.5 + \sin^2 x$ with the graph of $y = 1 + \cos 2x$ in the interval $0 \leqslant x \leqslant 2\pi$. **[3]**